VaultX Cyber Tech AI Internship
Week 7 Report

AI Final Project: Solar Installer Lead-to-Quote Automation (WhatsApp Delivery)

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Internship	VaultX Cyber Tech AI
Week	Week 7 (Final Project: AI Business Automation)
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1. Project Overview

The final project asked for a practical AI-powered automation solution for a real SME or local business in Pakistan, following the path Real Problem → AI Solution → Automation → Business Value. This report follows that path.

Project

SolarFlow: a WhatsApp lead-to-quote system for a small solar installation company. It turns an inquiry into an estimate and a PDF quotation within seconds, follows up automatically, understands the customer's replies, hands unusual messages to the owner, and sends a weekly report.

Real Problem

Customers send their inquiries on WhatsApp. The owner prepares each estimate by hand from the customer's electricity bill and appliances, and cannot follow up with everyone. Leads are forgotten, replies are slow, and because solar prices keep rising, a slow estimate is quickly out of date and the customer is lost.

AI Solution

An AI model reads the customer's WhatsApp message (English or Roman Urdu) and extracts the bill, city, appliances and language. A second AI step classifies later replies as interested, negotiating, not interested, a new estimate request, or something else. All calculations are done by code, not by the AI.

Automation

- Estimate, itemised cost, lead score (hot / warm / cold) and a PDF quotation are produced and sent automatically.
- Every lead is saved in a Google Sheet with its status, score, estimate and language.
- A scheduled workflow sends follow-up 1 and follow-up 2 to leads that have not replied, and stops once the customer replies.
- Replies update the lead status, receive a reply in the customer's language, and alert the owner by email; unknown messages are handed to the owner.
- A weekly report is emailed to the owner, and workflow errors are logged to an Errors sheet.

Business Value

- Instant estimates and quotations at any hour, without staff involvement.
- No lead is forgotten, because follow-ups run automatically.
- Prices live in one sheet (Price_List), so every new estimate uses the latest rates.
- The owner sees hot leads immediately and gets a weekly view of the pipeline.
- People are involved only where needed: price discussions, site visits, unusual questions.

Business Rules

Estimates use the monthly bill at an assumed 50 PKR per unit and 120 units per kW-month, with the system size rounded up to the nearest 0.5 kW. Only the panel price (40 PKR per watt) comes from a public source; the other rates are assumptions (see Section 3 and Section 6). A lead is hot at a bill of 40,000 or more, warm at 20,000 or more, otherwise cold.

Why this matters

Week 6 produced a booking agent that talks to the customer. Week 7 applies the same WhatsApp, n8n and "code for facts, AI for conversation" ideas to a full sales process: lead capture, quotation, follow-up, reply handling and reporting.

2. Tech Stack & Architecture

Tools and Technologies Used

- **n8n** (self-hosted, local): builds all five workflows; carried over from Weeks 3, 5 and 6.
- **OpenAI gpt-4o-mini:** extracts details from messages (Information Extractor) and classifies replies (Text Classifier).
- **GreenAPI:** sends and receives WhatsApp messages and files.
- **ngrok:** exposes the local n8n webhook to the internet so GreenAPI can reach it.
- **Google Sheets:** Leads, Price_List and Errors. **Google Docs and Drive:** quotation template and PDF export. **Gmail:** owner alerts and the weekly report.

Data Flow (WF1: Lead Estimate and Reply Handling)

```
WhatsApp → GreenAPI → ngrok → n8n Webhook → Filters (incoming text from the allowed number only)
→ Edit Fields → Get row(s) in sheet1 (look up the phone number) → Latest Row (newest lead) → Existing lead?
NEW CUSTOMER: Information Extractor → If1 (bill, city and a digit in the message?)
true: Get Price_List → Calculation (code) → Append row in sheet → Send Estimate → Prepare Quote Data → WF2 (PDF)
false: Send Estimate1 (asks for bill, city and appliances)
EXISTING CUSTOMER: Text Classifier → five branches:
interested / negotiating / not_interested: Update row → Reply → Notify Owner (Gmail)
new_estimate: back to Information Extractor (new estimate and new row)
other: Notify Owner (Gmail) → Send Other Reply
```

Other Workflows

Workflow	Purpose
WF2 Quote PDF	Called by WF1. Copies a Google Doc quotation template, fills in the customer's values, exports a PDF and sends it on WhatsApp with a caption in the customer's language.
WF3 Follow-ups	Schedule trigger every minute. Sends follow-up 1, then follow-up 2, to leads that have not replied, in their language, and updates their status.
WF4 Weekly Report	Every Monday: totals for the last 7 days (leads, hot / warm / cold, pipeline value, follow-ups) emailed to the owner.
WF5 Error Handler	Error workflow of the Lead Estimate workflow (WF1). Writes time, workflow name and error message of a failed run to the Errors sheet.

Design Principle: Code for Facts, AI for Conversation

This is the main lesson of Week 6, applied again. The AI only understands language (what the customer wrote, what kind of reply it is). System size, itemised cost, totals and scores are calculated by code, so the numbers are exact and repeatable. A code guard also stops the AI from guessing: if a message contains no number, no estimate is created.

3. Implementation Summary

Components Built

- Five n8n workflows: WF1 Lead Estimate, WF2 Quote PDF, WF3 Follow-ups, WF4 Weekly Report, WF5 Error Handler.
- A Google Sheet "SolarFlow" with three tabs: Leads, Price_List and Errors.

- A Google Doc quotation template and an AI extraction schema (bill, city, appliances, language).
- Code nodes for the calculation, the newest-lead lookup, the follow-up timing and message selection.

Node	Role
Webhook + Filters	Receive the WhatsApp message from GreenAPI; keep only real incoming text from the allowed number.
Get row(s) + Latest Row	Find this phone number in Leads and choose the newest row (sorted by created_at).
Existing lead?	Sends a known customer to reply handling and an unknown one to the estimate path.
Information Extractor	AI extracts bill, city, appliances and language from the message.
If1	Continues only if a bill, a city and at least one digit in the message are present; otherwise asks for details.
Get Price_List + Calculation	Reads the rates and calculates system size, itemised cost, total and lead score.
Append row / Update row	Saves a new lead; updates the status of an existing lead (only the changed columns).
Send Estimate / Send Estimate1	WhatsApp text in English or Roman Urdu, chosen by the detected language.
Prepare Quote Data + Call WF2	Builds the data for the quotation (including language) and calls the PDF workflow.
Text Classifier	AI sorts a customer's reply into interested, negotiating, not_interested, new_estimate or other.
Reply nodes	Send a fitting reply in the customer's language for each branch.
Notify Owner (Gmail)	Emails the owner: HOT LEAD, better price requested, not interested, or an unhandled message.

Estimate Logic (Code, Not AI)

```
units = bill / 50
system size (kW) = ceil((units / 120) x 2) / 2
total = panels + inverter + structure + installation + net metering
```

Item	Rate (PKR)	Source
Solar panels	40 per watt	Public price source (solarpanelprices.pk)
Inverter	25,000 per kW	Assumption
Mounting structure	8,000 per kW	Assumption
Installation	12,000 per kW	Assumption
Net metering	30,000 fixed	Assumption

Worked example: a bill of 30,000 gives 600 units and a 5 kW system. Cost: panels 200,000 + inverter 125,000 + structure 40,000 + installation 60,000 + net metering 30,000 = **Rs 455,000**. This is the value seen in the live WhatsApp test (Fig 3).

Reused from Previous Weeks

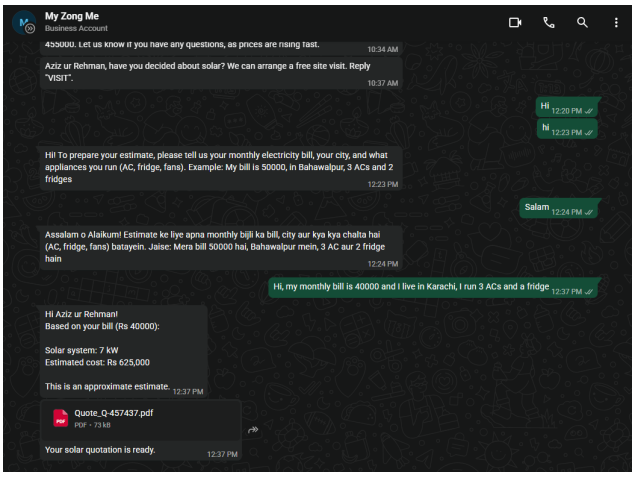
The n8n workflow habits from Week 3, the webhook and ngrok setup from Weeks 5 and 6, GreenAPI delivery and the "code over LLM" lesson from Week 6, and the secret-handling rule from Week 6 (the exported workflow files must carry the placeholder YOUR_API_TOKEN, never the real token).

4. Testing & Evidence

All tests were run through real WhatsApp messages from my own tester number, with the workflows published and running.

4.1 Estimate and PDF Quotation (English and Roman Urdu)

An English inquiry produces an English estimate and a PDF with an English caption; a Roman Urdu inquiry produces a Roman Urdu estimate and caption.



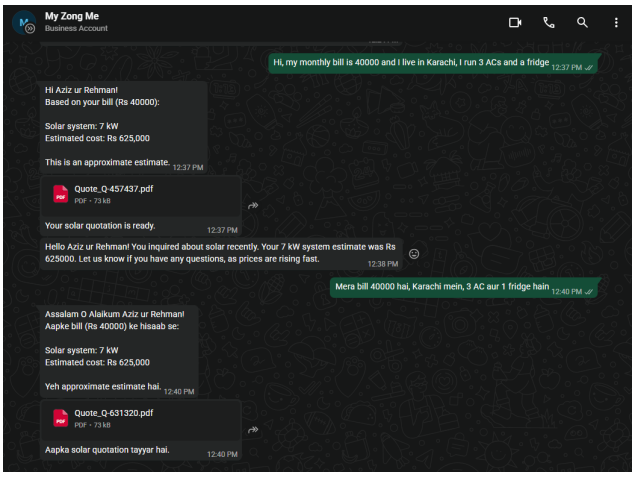


Fig 1 (left): English inquiry (bill 40,000): 7 kW estimate of Rs 625,000 and the PDF with an English caption. Fig 2 (right): after an English reply, a Roman Urdu inquiry gets a Roman Urdu estimate and caption.

4.2 Greetings Do Not Create Estimates

Before the guard was added, the single word "Hi" made the AI invent a bill and send a wrong estimate. Now "hi", "salam" and "hello" only receive a request for details, in the matching language; a proper message then receives the estimate.

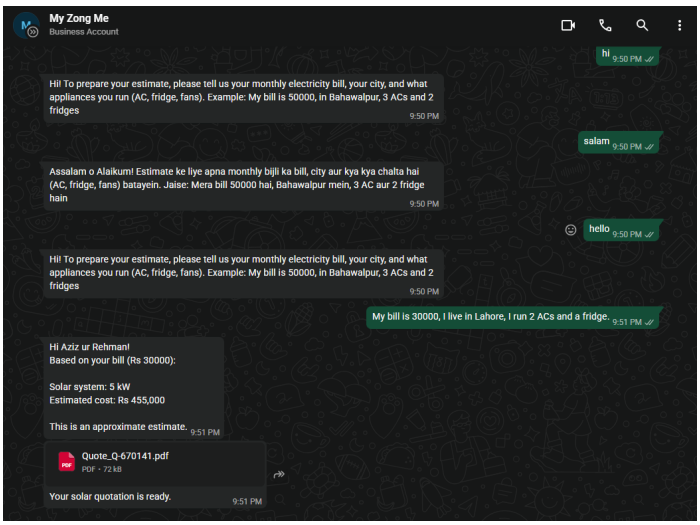


Fig 3: Greetings answered with a request for details; the full inquiry then returns a 5 kW estimate of Rs 455,000 and the PDF.

4.3 Follow-ups and Reply Handling

A lead that does not reply receives follow-up 1 and follow-up 2. When the customer replies, the AI classifies the message, the status is updated, the customer receives a reply and the owner is alerted. Fig 4 shows the three classified replies after follow-up 2.

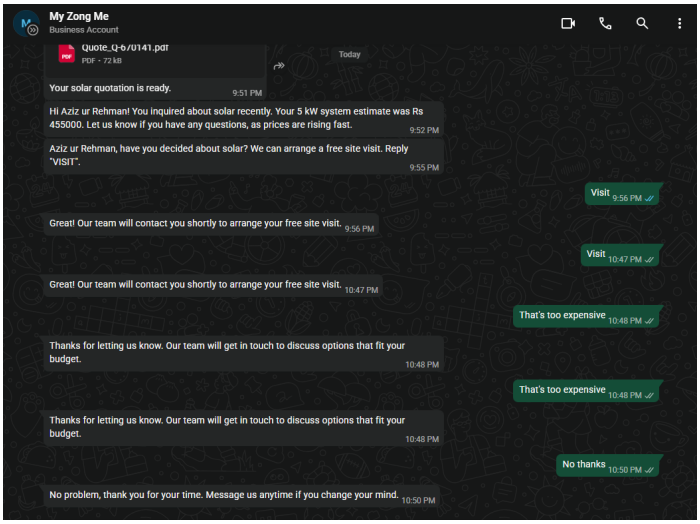


Fig 4: Follow-ups, then "Visit", "That's too expensive" and "No thanks" each answered with a matching reply.



Fig 5: The three reply branches in WF1: Update row → Reply → Notify Owner for interested, negotiating and not interested.

4.4 Owner Alerts and Weekly Report

Each branch sends the owner an email with the customer's name, phone, city, estimate and message. Unknown messages (for example "Where is your office?") are handed to the owner and the customer is told the team will contact them. The weekly report arrives as a separate email.

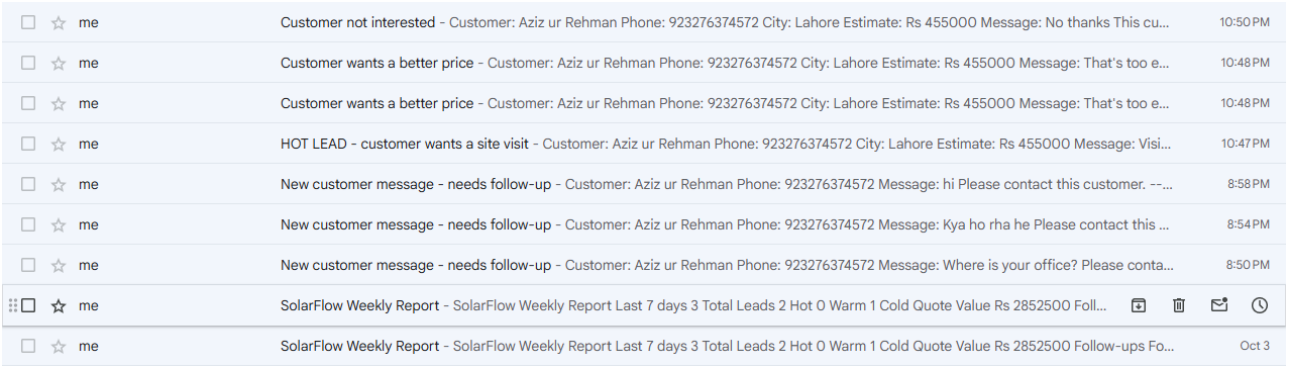


Fig 6: Owner inbox: not interested, better price, HOT LEAD and "needs follow-up" alerts, and the SolarFlow Weekly Report.

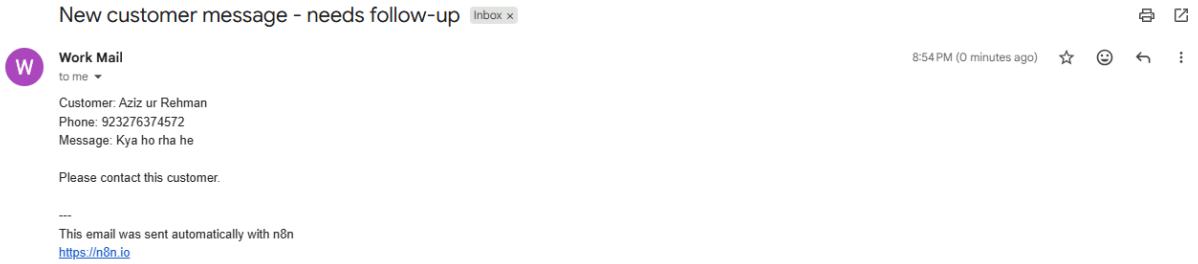


Fig 7: A customer message handed to the owner, with name, phone, message and instruction on separate lines.

4.5 Lead Data and Error Log

he	city	monthly_bill_pkr	load_details	system_size_kw	estimate_pkr	score	status	last_followup	language
923276374572	Lahore	30000	2 AC aur 1 fridge hain	5	455000	warm	followup2	2026-10-04T13:03:43.466+05:00	ur

Fig 8: Leads sheet row with phone, city, bill, estimate, score, status, last follow-up and the stored language.

time	workflow	error_message
2026-10-04T09:26:15.912+05:00	FinalProject-WF1 - Lead Estima	Wrong type: '1790937930339' is a number but was expecting a string [condition 0, item 0]
2026-10-04T12:20:58.670+05:00	FinalProject-WF1 - Lead Estima	The connection cannot be established, this usually occurs due to an incorrect host (domain) value

Fig 9: Errors sheet: two real workflow errors written automatically by the error workflow.

4.6 Test Summary

Test	Result
English inquiry: estimate, PDF and caption	Pass: 5 kW / Rs 455,000 for a bill of 30,000
Roman Urdu inquiry: estimate and caption	Pass: estimate and caption in Roman Urdu
Greetings (hi, salam, hello)	Pass after the guard fix: only a request for details is sent
Follow-up 1 and follow-up 2	Pass in English and Roman Urdu; replied leads are not followed up
Reply classification (interested, negotiating, not interested)	Pass: status updated, reply sent, owner emailed
Unrelated message ("Where is your office?")	Pass: owner emailed, customer told the team will contact them
Changed bill ("My bill is now 20000")	Pass: new estimate, PDF and new lead row
Weekly report email	Pass: received by the owner
Error handler	Pass: real errors written to the Errors sheet
Reply updates the newest row when one number has several rows	Fix applied; a clean retest with two rows is still pending

5. Refinements After Initial Testing

Live testing exposed several issues. Each one is recorded below with its cause and fix.

5.1 The AI Invented an Estimate for "Hi"

A plain "Hi" produced a bill of Rs 3,500 and an estimate, because the extractor guessed values that were not in the message. **Fix:** the bill attribute now says to leave it empty and never guess, and the If1 node has a third condition that requires a digit in the customer's message, so the guess cannot pass. The model cannot be trusted alone; code makes the final decision.

5.2 Replies from Existing Customers Were Treated as New Inquiries

A returning customer received the "send your bill" message. The lookup used the WhatsApp ID with the @c.us suffix, but the sheet stores the number without it, and the IF compared a numeric lead_id with a string. **Fix:** the suffix is removed before the lookup and "Convert types" is enabled in the condition.

5.3 Several Rows for One Phone Number

A customer who asks for a new estimate gets a new row, and the first lookup always returned the oldest. **Fix:** the lookup now returns all matches and a code node sorts them by created_at and passes on the newest row.

5.4 Updates Could Blank Other Columns

After a refresh, the Update row node showed an extra empty "language" field. An empty mapped field can overwrite the stored value. **Fix:** the unused fields were deleted so only lead_id and status are sent.

5.5 A Wrong Connection Created a Loop

The new_estimate output was first connected back into the Text Classifier input, which would make the workflow run itself again. **Fix:** the line was removed and connected to the Information Extractor. Execution screenshots made the mistake visible.

5.6 No Messages and One-Line Emails

Outputs of some branches did not reach the customer because a branch had no connection, and the owner email ran all lines together. **Fix:** every branch was connected to a reply node, and the email text is built with explicit line breaks.

5.7 Network and Credential Errors

A message failed once with "connection cannot be established" and a Google column list stopped loading; both were temporary or a credential problem, solved by retrying and reconnecting the Google account. The error workflow logged the first one.

5.8 Secret Handling

As learned in Week 6, secrets are never committed. Exported workflow files must contain the placeholder YOUR_API_TOKEN, and the real token stays in n8n.

6. Known Limitations

These issues were found and are documented rather than hidden:

- **No real owner interview:** I could not reach a solar installer. Rates other than the panel price, the 50 PKR per unit and 120 units per kW-month figures, the lead-score thresholds, the company name and the quotation terms are assumptions or placeholders.
- **Estimate accuracy:** sizing is a rough estimate; a real installer would also consider roof area, sunlight and load profile.
- **Greetings from a known customer:** a message such as "hi" from an existing customer goes through the classifier and is treated as "other", so the owner receives an email.
- **Questions from brand-new customers:** they receive the "send your bill" message, not an answer, and the owner is not alerted.

- **Language of replies:** it comes from the language stored for the lead, not from each new message, so an English question from a Roman Urdu lead gets a Roman Urdu reply.
- **Number words:** the digit guard misses amounts written in words, for example "thirty thousand".
- **WhatsApp free plan:** only a few chats are allowed, so only my tester number passes the filter.
- **PDF template:** it is in one language; the caption and the messages are language-aware.
- **Demo timing:** follow-ups use short minute values for the demo; real use would be one day and three days.
- **Local setup:** n8n and ngrok run on a local machine, and the free ngrok URL changes after a restart, so the GreenAPI webhook URL must be updated each time.

7. Conclusion / Learnings

This week connected the earlier WhatsApp and n8n work into a complete sales workflow: it creates an estimate and a quotation, keeps the lead warm, understands replies, and involves the owner only when a person is needed. The clearest lesson came from the "Hi" bug: even a well-instructed AI can invent data, so every decision that matters needs a code check next to it.

Three practical lessons will save time on future workflows: look up a record before deciding what a message means; when one key can have several rows, choose the newest deliberately; and map only the columns you intend to change.

Finally, the open items in Section 6 are listed on purpose. A working system with honest, documented limits is more useful than one that claims everything is finished.